Spin-1 dimer and trimer singlets

Exact quantum construction and first validation

Why spin 1 is the direct choice

Both two and three spin-1 particles have a unique total-spin-zero state.
Use normalized dimers and trimers with equal covering coefficients.
The trimer is antisymmetric; its sites are ordered x, y, z on every face.

The classical result does not transfer

Different singlet coverings overlap. A single cube move between
eight trimers and twelve dimers gives overlap 1/39,366.
An exact tensor network with virtual dimension four includes these overlaps.

What was actually calculated

All 8,192 coverings on a periodic 2 × 2 × 2 cluster of 24 spin-1 sites.
Norm and physical singlet-projector correlations use exact arithmetic.
Pair singlet expectation: 0.99354414. Face singlet: 0.03915282.
The reported quantum contractions took 3.22 seconds.

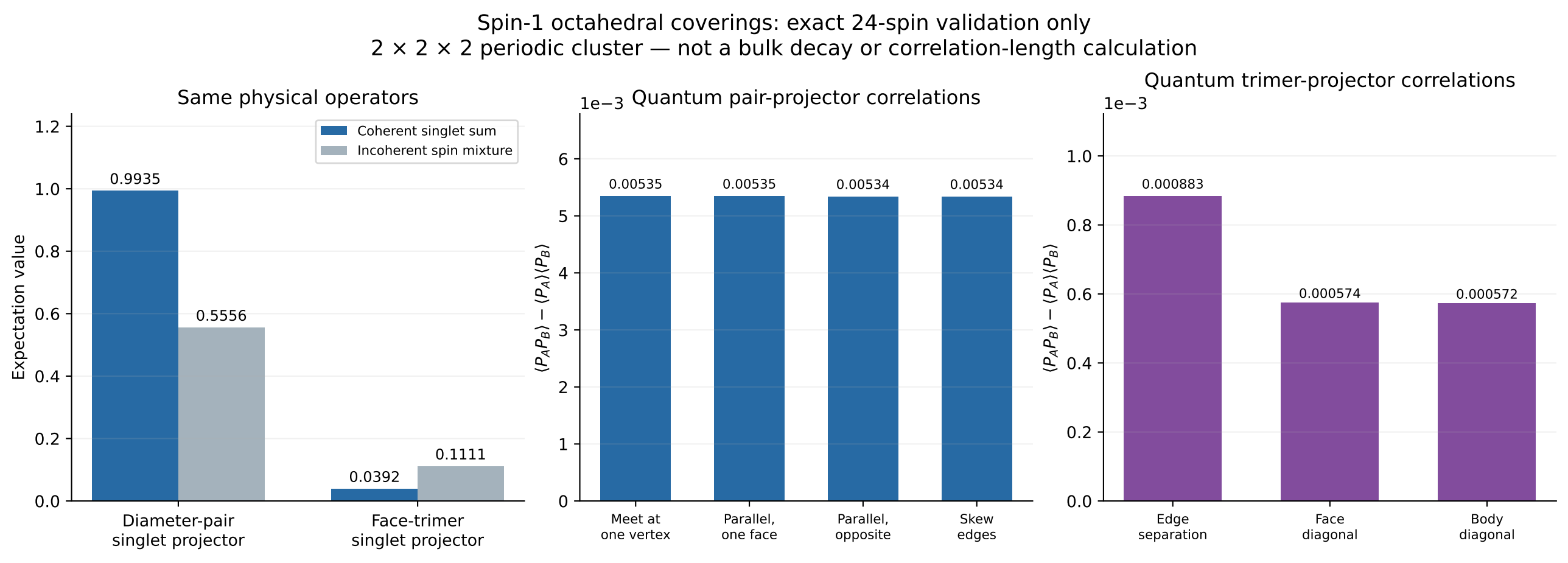
Essential finite-size limitation

On this torus all 4,096 dimer coverings become the same physical state.
Their coherent enhancement strongly affects the measured values.
These are construction checks. The bulk quantum correlation length
and exponential-versus-power-law decay remain undetermined.

What spin one-half would require

Three spin-one-half particles have two doublets and one quartet,
but no singlet. Choosing trimer spin 1/2 or 3/2 leaves an effective spin.
Its internal state and coupling to other trimers would need specifying.
That defines another quantum model, not an automatic simplification.

Reproduce: python scripts/run_octahedral_spin1.py
Exact fractions, source hashes and method details accompany this PDF.


Spin-1 octahedral coverings: exact 24-spin validation only
2 × 2 × 2 periodic cluster — not a bulk decay or correlation-length calculation
Same physical operators
Coherent singlet sum
Incoherent spin mixture
Expectation value
0.9935
0.5556
0.0392
0.1111
Diameter-pair singlet projector
Face-trimer singlet projector
Quantum pair-projector correlations
1e−3
$\langle P_A P_B \rangle - \langle P_A \rangle \langle P_B \rangle$
0.00535
0.00535
0.00534
0.00534
Meet at one vertex
Parallel, one face
Parallel, opposite
Skew edges
Quantum trimer-projector correlations
1e−3
$\langle P_A P_B \rangle - \langle P_A \rangle \langle P_B \rangle$
0.000883
0.000574
0.000572
Edge separation
Face diagonal
Body diagonal